DEPIN ARCHITECTURE OVERVIEW

The Node Operator Manifesto

Monetize Your Idle Hardware. Secure Enterprise Assets. Earn Decentralized USDT Payouts.

Target Audience Data Centers, Miners & Web3 Enthusiasts

Settlement Protocol BNB Chain Smart Contracts

Payout Currency USDT (Stablecoin)

Network Status Active Testing Phase

DEPIN ARCHITECTURE

Sovereign Data Storage

Powered by DePIN — encrypted, sharded, and settled across a distributed swarm of independent node operators.

SECTION 01

Why Join the Swarm?

> Stop letting your surplus storage capacity depreciate. The Inaya Network transforms your idle drives into cash-generating enterprise infrastructure, protected by trustless mathematics.

Traditional cloud providers like AWS and Google Cloud monopolize the $200B+ data storage industry. The Inaya Network (DePIN) disrupts this by decentralizing the physical hardware layer. We provide the cryptographic software, the zero-knowledge frontend, and the corporate clients. You provide the hard drive space.

By becoming an Inaya Node Operator, you join a global distributed fleet that stores heavily encrypted, mathematically sliced data shards. In exchange for your hardware's storage and bandwidth, you earn consistent USDT revenue transferred to your Web3 wallet via our automated settlement smart contracts.

STABLECOIN PAYOUTS

Unlike other networks that pay in volatile native tokens, your core storage capacity commissions are settled strictly in USDT, ensuring predictable ROI.

ZERO-KNOWLEDGE LIABILITY

You only store encrypted binary shards. Because you never hold complete files, you have absolute zero legal liability regarding the data contents you host.

NO EXPENSIVE RIGS REQUIRED

Forget GPU mining. You don't need expensive compute hardware. A standard server or PC with a strong internet connection and reliable HDD/SSD space is enough.

SECTION 02

Tiered Commission Architecture

The Inaya Protocol enforces a strict, merit-based commission structure. Your payout percentages scale directly with the amount of storage you allocate and the reliability of your machine. The more you provide, the higher your cut of the corporate revenue.

PROVIDER TIER	HARDWARE REQUIREMENT	COMMISSION PAYOUT	NETWORK ROLE
Tier 1: Entry	Under 500 GB	30% of Revenue	Redundancy Swarm Node
Tier 2: Mid-Level	500 GB to 4.99 TB	40% of Revenue	Core Retrieval Node
Tier 3: Enterprise	5 TB and Above	50% of Revenue	Institutional Guardian Node

The 90% Uptime Gate (Quality Control)

Enterprise clients require reliable access to their data shards. To protect the network's reputation, our Smart Contract includes an unyielding Uptime Filter.

Downgrade Condition — If your node's connection drops and your weekly uptime score falls below 90.00%, the contract will automatically downgrade your payout to the baseline Entry Tier (30%), regardless of how many terabytes you are storing.

Restoration Condition — Once your hardware stabilizes and hits the 90%+ benchmark in the next cycle, the contract automatically restores your premium commission rate.

SOLIDITY — TIER CALCULATION LOGIC

```
function _calculateTier(uint256 _capacityGB, uint256 _uptimeScoreBps) internal view returns (Tier) {
    // Rigid Uptime Enforcement Gate
    if (_uptimeScoreBps < 9000) return Tier.Entry;

    if (_capacityGB >= 5000) return Tier.Enterprise;
    if (_capacityGB >= 500) return Tier.Mid;

    return Tier.Entry;
}
```

SECTION 03

Initialize Your Node

The Inaya ecosystem is designed for frictionless onboarding. Commissions are calculated automatically and released to your wallet without any action on your part — typically within 36 hours, giving the network a brief security window before funds move.

1. **Install the Inaya Node Daemon —** Windows, macOS, or Linux. No Docker required.
2. **Register your node —** Set your storage capacity and connect your wallet.
3. **Start earning —** The daemon runs in the background and reports uptime automatically.

Automated Batch Settlement

Our backend coordinator aggregates node telemetry over a 7-day period. At the end of the settlement cycle, the coordinator queues a secure, gas-optimized batch transaction on the BNB Chain.

Thanks to our robust smart contract engineering, even if an individual operator's queued settlement encounters an error, the system safely skips and logs the fault — every other node operator's payout still proceeds on schedule.

Mainnet Assignment Priority

Operators who maintain 95%+ uptime for at least 90 consecutive days on the testnet Watcher Program earn priority access when mainnet node assignments open.

New: Contribute to the Security Layer

The same node-daemon you're already running now supports a second, independent way to contribute: inaya-node-daemon report <indicator> submits a signed observation to Inaya's decentralized threat-intelligence network — the same reputation-weighted, on-chain-confirmed system that powers the public Inaya Firewall. It's a separate track from storage commissions, using the identity you've already registered.

> *Ready to deploy? Install the Inaya Node Daemon and register your wallet to begin.*